H2. Aziza Iklasova IT-2402 Checkpoint 3 TO3

$\min z = 3y_1 + 4y_2 + 5y_3$

s.t. $y_1 + y_2 \geq 1$, $y_2 + y_3 \geq 1$, $y_1 + y_3 \geq 1$

$y_1, y_2, y_3 \in \{0,1\}$

LP relaxation: $0 \leq y_1, y_2, y_3 \leq 1$

a) $y_1 = y_2 = y_3 = \frac{1}{2}$

$y_1 + y_2 \geq 1$ : $\frac{1}{2} + \frac{1}{2} = 1 \geq 1$ ✓

$y_2 + y_3 \geq 1$ : $\frac{1}{2} + \frac{1}{2} = 1 \geq 1$ ✓

$y_1 + y_3 \geq 1$ : $\frac{1}{2} + \frac{1}{2} = 1 \geq 1$ ✓

$0 \leq \frac{1}{2} \leq 1$ ✓ All variables satisfy LP bounds

$y = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ is LP feasible

$z = 3y_1 + 4y_2 + 5y_3 = 3 \cdot \frac{1}{2} + 4 \cdot \frac{1}{2} + 5 \cdot \frac{1}{2} = 6$

The upper bounds $y_i \leq 1$ can be omitted without changing the opt. value, because all costs are positive. The dual of the resulting covering LP:

$\max \pi_1 + \pi_2 + \pi_3$

s.t. $\pi_1 + \pi_3 \leq 3$

$\pi_1 + \pi_2 \leq 4$

$\pi_2 + \pi_3 \leq 5$

$\pi_1, \pi_2, \pi_3 \geq 0$

$\pi_1 + \pi_3 = 1 + 2 = 3 \leq 3$

$\pi_1 + \pi_2 = 1 + 3 = 4 \leq 4$

$\pi_2 + \pi_3 = 3 + 2 = 5 \leq 5$

all $\pi_i \geq 0$, dual point is feasible

obj. value: $b^T \pi = 1 \cdot 1 + 3 \cdot 1 + 1 \cdot 2 = 6$

weak duality: $\underbrace{b^T \pi}_{\text{dual lower bound}} \leq z_{LP} \leq \underbrace{z(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})}_{\text{primal feasible cost}}$

$6 \leq z_{LP} \leq 6$, $z_{LP} = 6$

point $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ is feasible and optimal for the LP relaxation with objective cost 6

b) $y = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$

Round down $y^{down} = (0,0,0)$

$T_1$: $0+0+0 \not\geq 1$, fail

$T_2$: $0+0 = 0 < 1$, fail

$T_3$: $0+0 = 0 < 1$, fail

$z = 3 \cdot 0 + 4 \cdot 0 + 5 \cdot 0 = 0$

cost = 0, infeasible the cost is 0 we cannot accept solution because it violates all 3 constraints.

Round UP $y^{up} = (1,1,1)$

$T_1$: $1+1 = 2 \geq 1$ ✓

$T_2$: $1+1 = 2 \geq 1$ ✓

$T_3$: $1+1 = 2 \geq 1$ ✓

$z = 3 \cdot 1 + 4 \cdot 1 + 5 \cdot 1 = 12$, feasible

Mixed roundings (the different combinations of zeros and ones)

| $y = (y_1, y_2, y_3)$ | $T_1$ | $T_2$ | $T_3$ | feasible? | cost |
|---|---|---|---|---|---|
| (0,0,0) | fail | fail | fail | no | 0 |
| (1,0,0) | pass | fail | pass | no | 3 |
| (0,1,0) | pass | pass | fail | no | 4 |
| (0,0,1) | fail | pass | pass | no | 5 |
| (1,1,0) | pass | pass | pass | yes | 7 |
| (1,0,1) | pass | pass | pass | yes | 8 |
| (0,1,1) | pass | pass | pass | yes | 9 |
| (1,1,1) | pass | pass | pass | yes | 12 |

the cheapest feasible combination is

$y^* = (1,1,0)$

$z^* = 3 \cdot 1 + 4 \cdot 1 + 5 \cdot 0 = 7$

c) find the integer optimum and integrality gap

1. state both opt.values

$z_{LP} = 6$ $z^* = 7$, the LP relaxation has a lower cost because it permits fractional variables

2. calculate integrality gap

$$Gap = \frac{z^* - z_{LP}}{z^*} = \frac{7-6}{7} = \frac{1}{7} \approx 0.1429 \approx 14.29\%$$

3. why no integer point can be optimal for the LP

we proved $z_{LP} = 6$, the cheapest feasible binary solution costs: $z^* = 7$
every feasible binary solution has cost of least 7: $z_{binary} \geq 7$,
no feasible binary point can achieve the LP bit of 6 $\Rightarrow$
No integral point is optimal for the LP relaxation, which shows a positive integrality gap.